

Database Systems (CS-151)

WEEK-01

Course Instructor: Munazza

Course administration

Instructor: Munazza

(Lecturer, Deptt of Computer Science, University of Wah, Wah Cantt)

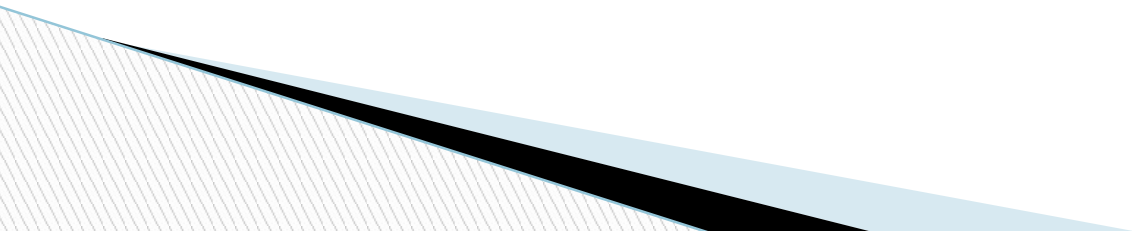
Contact Information:

munazza.bibi@uow.edu.pk

For queries related to course(Assignments/Projects) just leave an email at

munazza.bibi@uow.edu.pk

Consultation Hours 02:00PM - 03:00 PM



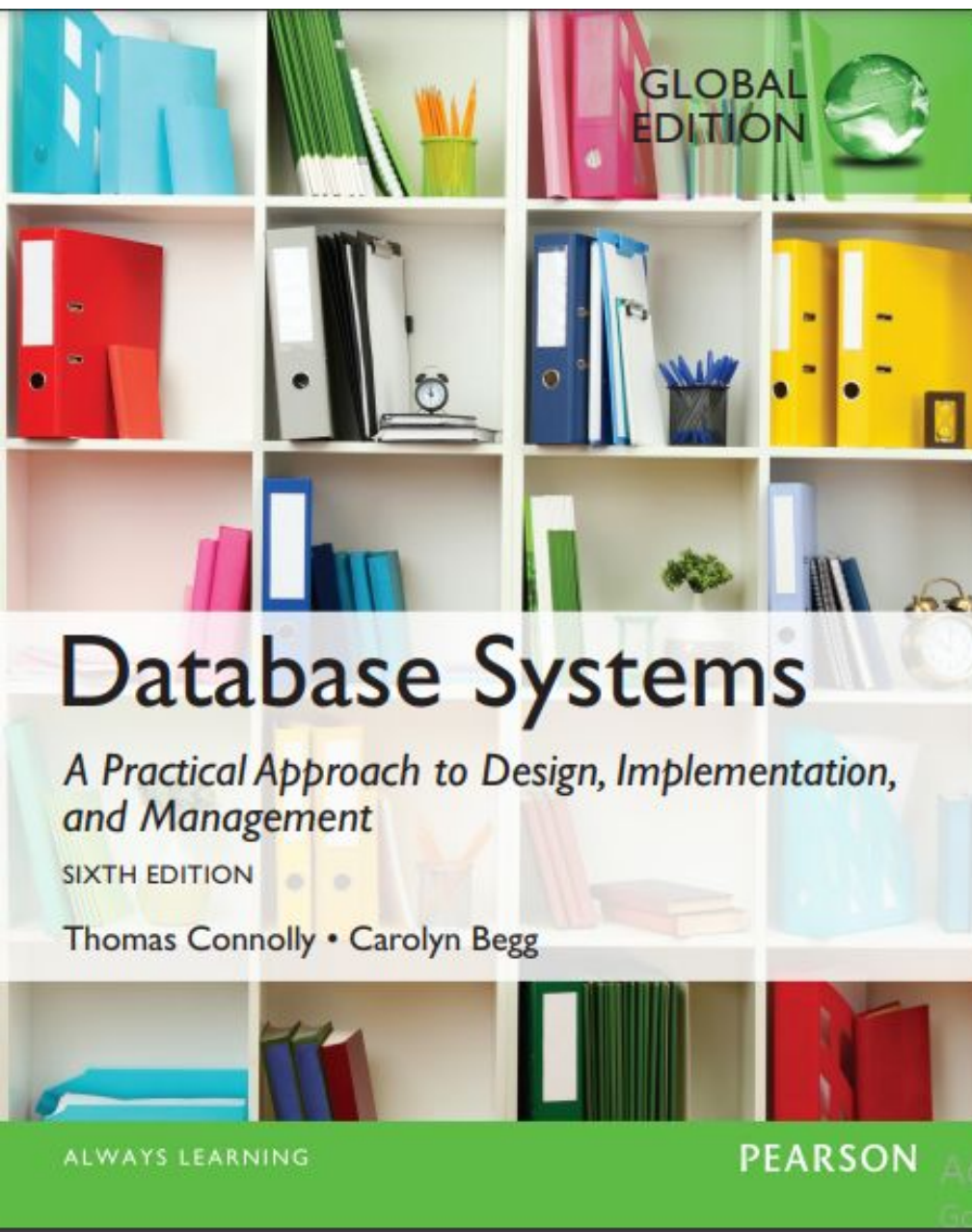
Course Learning Outcomes

Course Learning Outcomes		
After successful completion of this course, the students should be able to:	GA	BT Level
1. Explain fundamental database concepts	2	C2
2. Identify functional dependencies and Produce SQL queries for database definition and manipulation in any DBMS	3	C3
3. Discover conceptual, logical and physical database schemas using different data models	4	C4

Mapping of GA's/PLO's with CLO's

Mapping of CLOs to GA's

GA's/CLOs	CLO1	CLO2	CLO3	CLO4
GA1 (Academic Education)				
GA2 (Knowledge for Solving Computing Problems)	✓			
GA3 (Problem Analysis)			✓	
GA4 (Design/Development of Solutions)		✓		
GA5 (Modern Tool Usage)				
GA6 (Individual and Teamwork)				
GA7 (Communication)				
GA8 (Computing Professionalism and Society)				
GA9 (Ethics)				
GA10 (Life-long Learning)				



Recommended Textbook and Reference Books

Text Book:

Database Systems: A Practical Approach to Design, Implementation, and Management by Thomas Connolly and Carolyn Begg 6th Edition, 2021

Text Book:

Database Management Systems by Raghu Ramakrishnan, Johannes Gehrke, 3rd Edition 2002 Database System Concepts by Avi Silberschatz, Henry F. Korth and S. Sudarshan, 7th Edition, 2019

Database Systems: The Complete Book by Hector Garcia-Molina, Jeffrey D. Ullman, Jennifer Wido, 2nd Edition, 2008

Course Overview

- ◆ **Introduction and significance of Databases**
- ◆ **Database Environment**
- ◆ **Database Languages**
- ◆ **Database Architectures and the web**
- ◆ **Relational model**
- ◆ **Database system development lifecycle**
- ◆ **Entity-relationship modelling**
- ◆ **Structural constraints**
- ◆ **Enhanced Entity-relationship modelling**
- ◆ **Normalization**
- ◆ **Functional dependencies**
- ◆ **SQL: Data Manipulation**
- ◆ **SQL Queries**
- ◆ **Ethical and security issues in database**

Pre-requisites

◆ None

Grading criteria

Assessment

%Age

Assignments/ Quizzes

18.75%

Viva/Sessional

25%

Mid Term Exam

22.5%

Final Term Exam

33.75%

Total

100%

Goals for today

◆ To introduce:

- Introduction to database
- Traditional file processing system
- Advantages/Disadvantages of database

Data

- ❖ A collection of raw facts and figures related to an object
 - ❖ Object may be a person, an organization, an event, or any other thing that is significant in a system
- ❖ Data may be in form of text, numbers, images, sounds, and videos
- ❖ Collected for different purposes
- ❖ Processed to produce meaningful information
 - ❖ reports, charts, and web pages, etc.

Example

- ❖ May be collected to prepare the result of an examination of students
- ❖ To completely understand it must be processed according to requirements in a system

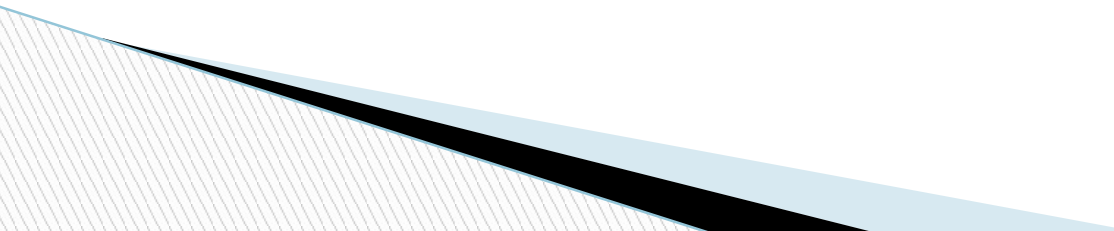
Roll No	Name	English	Chemistry	Computer
1	Saleem	62	63	64
2	Babar	50	75	70
3	Amanat	90	80	70
4	Salma	75	80	60
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Information

- ❖ Usually, collected data is not in proper format and does not give proper meanings
- ❖ Various operations are performed to get the required results that give proper and useful meanings known as information
- ❖ Information means processed data
 - ❖ Then used for decision-making, analytical purposes, and fault diagnosis, etc.
- ❖ Data is processed using various techniques



Manual File-Based System

- ❖ Oldest system used for records keeping in an organization
 - ❖ A set of books (or files) is prepared that contains a particular set of information
 - ❖ Each department of the organization has a separate file (or set of files) for every significant task
 - ❖ Information is shared among different departments through files
 - ❖ Many such files are labeled and placed in one or more cabinets
 - ❖ For Security, cabinets are locked or may be located in secure areas of the building
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Manual File-Based System

- ◆ **Example:**
- ◆ College/University manual file-based system for maintaining records of students and courses of different faculties etc.
 - ◆ Students record file used to store records of each student
 - ◆ Students fee file used to store fee records of students
 - ◆ Students result file used to store the result of examinations of students
 - ◆ Students course file used to store data of courses taken by students

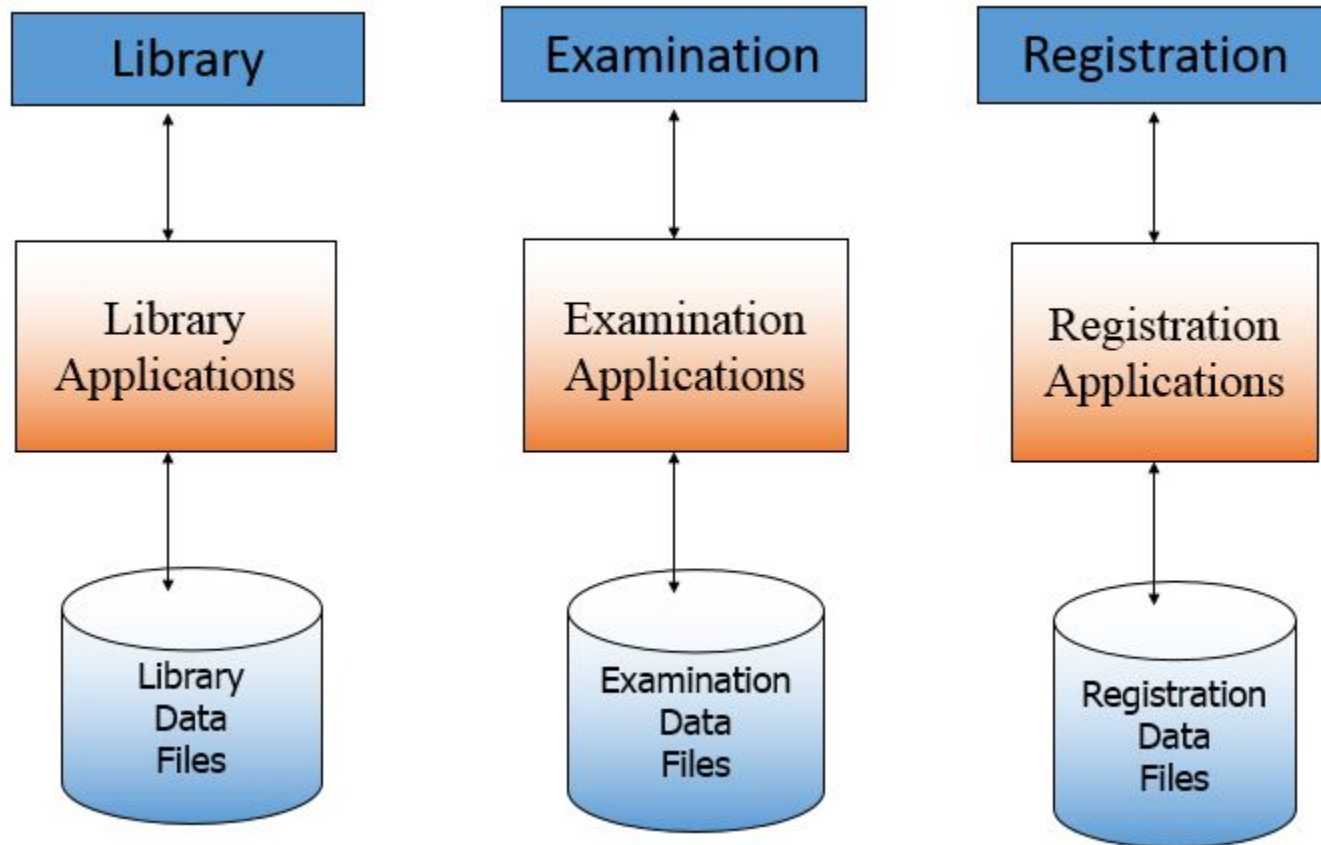
Disadvantages

- ❖ **Decentralized approach is adopted**
 - where each department processes and controls its own data
- ❖ **Very slow method to process data**
 - because data is analyzed and computed manually
 - data is transferred from one department to another manually
- ❖ **Very costly method**
 - because large number of employees have to be appointed to operate the file-based system
- ❖ **Difficulty in taking decisions**
- ❖ **Large space is needed to store the files/books**
 - Also difficult to search a specific file or a piece of information
- ❖ **Duplication of data may exist throughout the organization**

File Processing System

- ❖ It is also known as *Computer file processing system* or *traditional file processing system* or *computer file-based system*
- ❖ In early days of computer when there is no databases
 - ❖ Data was stored in computer files on tape or disk
 - ❖ Data was stored and managed through application programs
- ❖ Still used in some small organizations
- ❖ Each department has its own set of data files
- ❖ Each application program is developed with its own set of data files that is likely to be a subset of the master file

Traditional file processing environment



Program and Data Interdependence

Disadvantages of File Processing System

These subsets of the master file lead to

◆ **Data redundancy**

- Each application has its own data file so, same data may have to be recorded and stored in many times.

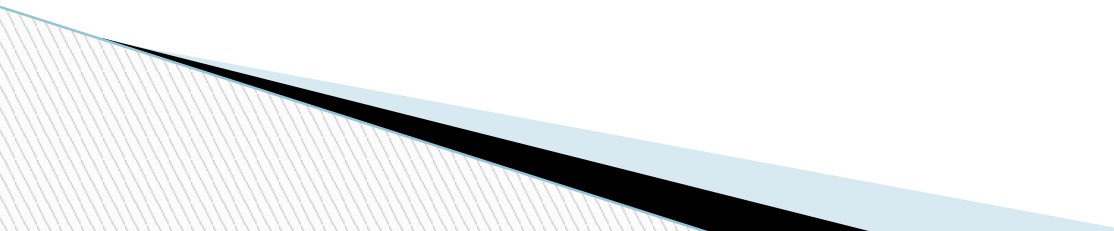
◆ **Data inconsistency**

- Due to the same data items that appear in more than one file do not get updated simultaneously in each and every file.

◆ **Data dependence**

- Program and application in the file processing system are data dependent but, the problem is incompatible with file format. For example, suppose data about the customer bills is stored in the file, and different programs use this file for different purposes, like adding data into the bills file, to compute the bill and to print the bill.
- Now the company asks to add the customers' address in the bills, for this we have to change the structure of the bill file and also the program that prints the bill.

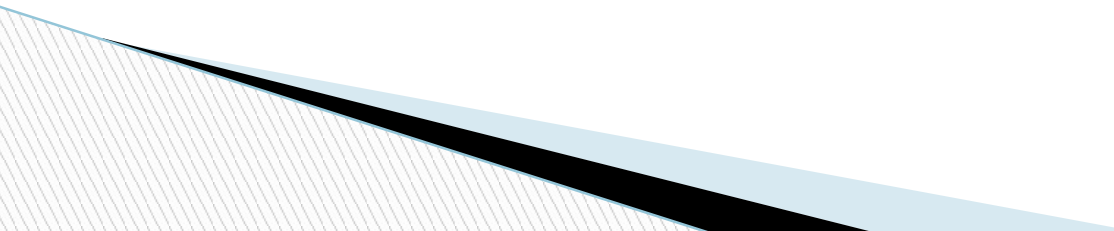
Disadvantages of File Processing System

- ❖ Limited data sharing.
 - ❖ The problem with security.
 - ❖ Retrieval (retrieval is not easy).
 - ❖ Time-consuming.
 - ❖ Inefficient to maintain the record of the big firm having a large number of items.
 - ❖ Required Lots of labor work to do.
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Database

- ◆ *“Database is an organized collection of related data stored in an efficient and compact manner”.*
- ◆ **"organized"** means that data is stored in such a way that it can easily be accessed and updated
- ◆ **"related data"** means data and information about a particular area is stored such as:
 - Database of employees contains data of employees of that particular organization or department
 - Database of students contains data of students of a college/university etc.
- ◆ **"efficient"** means required data can be searched very easily and quickly
- ◆ **"compact"** means that stored data takes up as little space as possible without any duplication of data

Examples of Databases

- ◆ NADRA
 - ◆ Library
 - ◆ College/University
 - ◆ Bank Accounts
 - ◆ E-mail Accounts
 - ◆ Etc.
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Types of databases

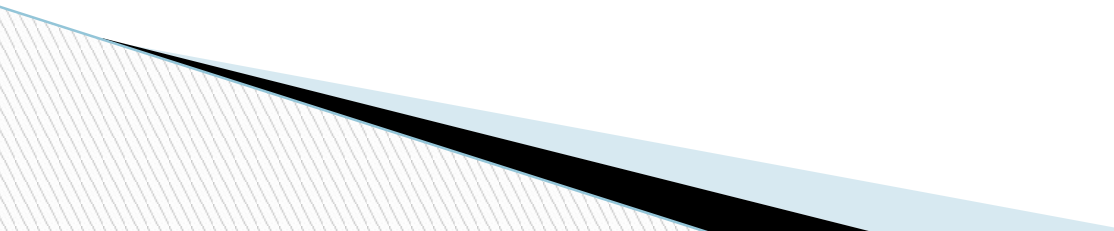
❖ Centralized databases

Database whose complete data is located at a single computer such as Personal computer database, client-server database and central computer database.

❖ Distributed databases

A single logical database which is spread physically across multiple computers over wide area network. Such databases are managed by the organizations having offices in different cities or countries.

Database Management System (DBMS)

- ❖ A collection of programs that are used to create, maintain, and extract data from databases
 - ❖ It is general-purpose software
 - ❖ This is often called database software
 - ❖ Different types of DBMSs are available, ranging from small systems that run on personal computers to huge systems that run on mainframes
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Functions of DBMS

❖ **Defining the Structure of Database**

- It involves defining tables, fields and their data types, constraints for data to be stored in the database, and the relationships among tables.

❖ **Populating the Database**

- Means storing data into database

❖ **Manipulating the Database**

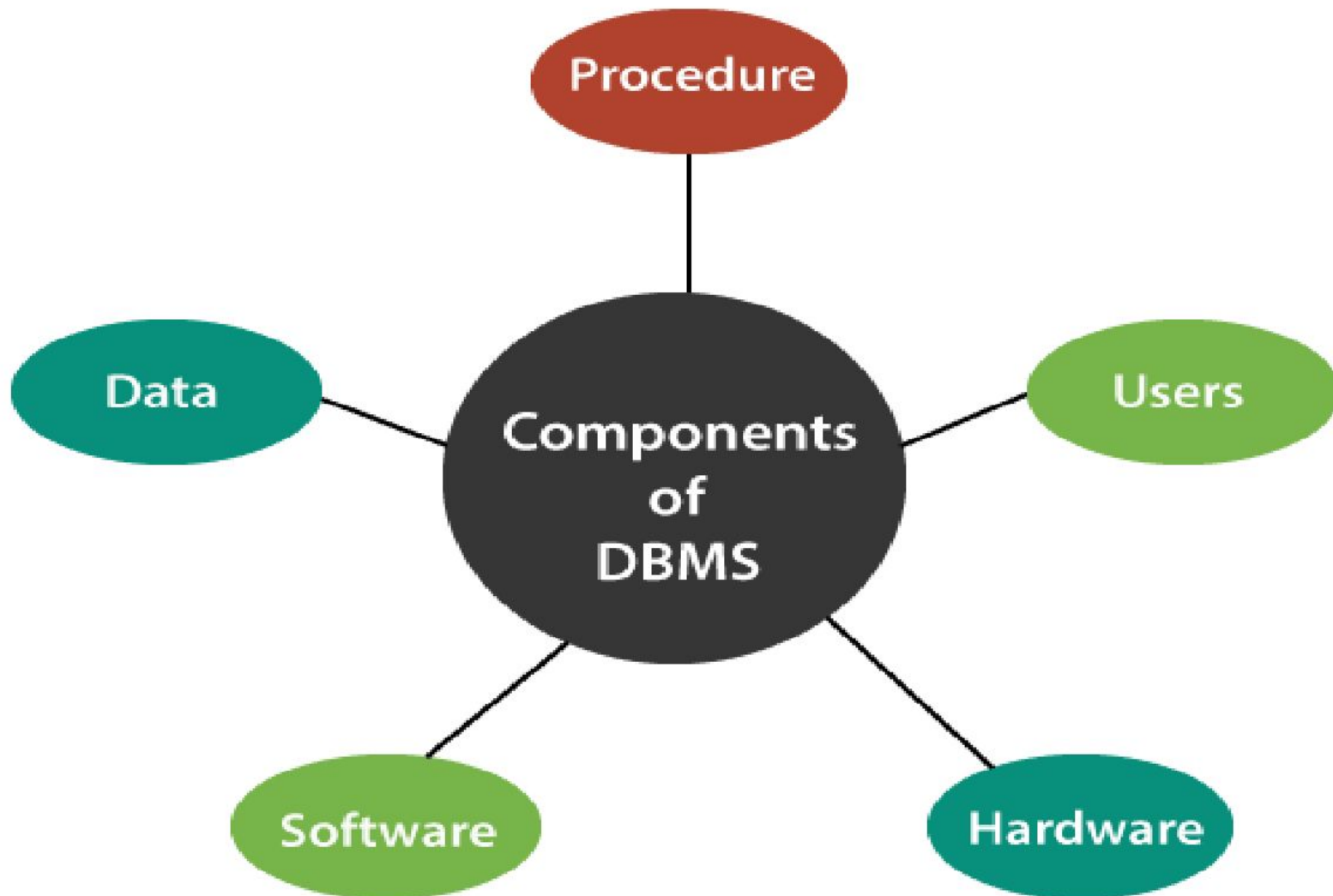
- In involves to retrieve specific data, update data, delete data, insert new data, and generate reports.

Examples of DBMS

- ❖ Microsoft Access
- ❖ Oracle
- ❖ Microsoft SQL Server
- ❖ MySQL
- ❖ FileMaker Pro



Components of a DBMS Environment



Components of a DBMS Environment

❖ Hardware

- ❖ This component of DBMS consists of a set of physical electronic devices such as computers, I/O channels, storage devices, etc. that create an interface between computers and the users.
- ❖ This DBMS component is used for keeping and storing the data in the database.

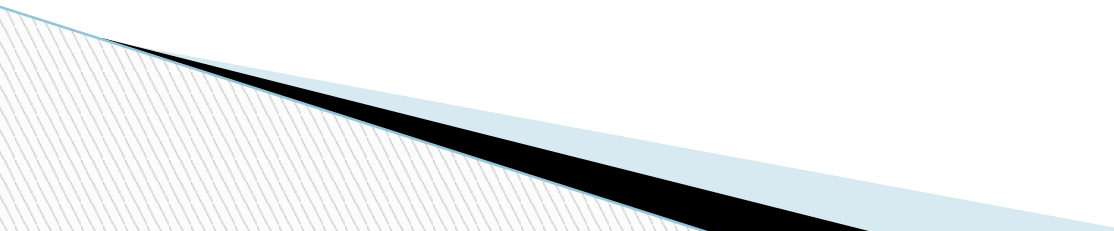
Components of a DBMS Environment

❖ Software

- ❖ The main component of a DBMS is the software.
- ❖ It is the set of programs which is used to manage the database and to control the overall computerized database.
- ❖ The DBMS software provides an easy-to-use interface to store, retrieve, and update data in the database.
- ❖ This software component is capable of understanding the Database Access Language and converts it into actual database commands to execute or run them on the database.

Components of a DBMS Environment

◆ **Data**

- ◆ It is the most important component of the database management system.
 - ◆ In DBMS, databases are defined, constructed, and then data is stored, retrieved, and updated to and from the databases.
 - ◆ The database contains both the metadata (description about data or data about data) and the actual (or operational) data.
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Components of a DBMS Environment

❖ Procedures

- ❖ Procedures refer to general rules and instructions that help to design the database and to use the DBMS.
- ❖ Procedures are used to;
 - ❖ setup and install a new DBMS
 - to login and logout of DBMS software
 - to manage DBMS or application programs
 - to take backup of the database
 - to change the structure of the database, etc.

Components of a DBMS Environment

◆ Users

- ◆ The users are the people who control and manage the databases and perform different types of operations on the databases in the database management system.
- ◆ There are three types of user who play different roles in DBMS:
 - Application Programmers
 - Database Administrators
 - End-Users

Different Users Role in DBMS

◆ **Application Programmers**

- ◆ The users who write the application programs in programming languages (such as Java, C++, or Visual Basic) to interact with databases are called Application Programmer.

◆ **Database Administrators (DBA)**

- ◆ A person who manages the overall DBMS is called a database administrator or simply DBA.

◆ **End-Users**

- ◆ The end-users are those who interact with the database management system to perform different operations by using the different database commands such as insert, update, retrieve, and delete on the data, etc.

Advantages of Database

◆ Controlling Data Redundancy

- In traditional file system, each application program has its own data files and this result in duplication of data in more than one places.
- Database approach reduces data redundancy.
- The data appears in a database appears only once and it is not duplicated.
- Through controlling data redundancy storage space is saved.

◆ Data Consistency

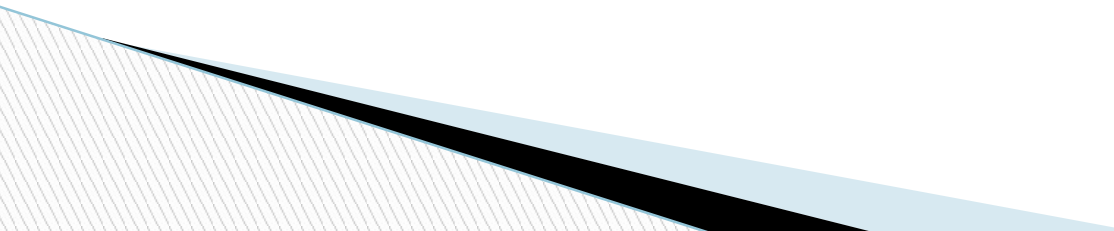
- By controlling data redundancy, data consistency is obtained.
- Data item is appeared only once
- Any update in the value is performed only once and it is readily available to everyone.

Advantages of Database

◆ Data Sharing

- Allows to share (access) data by any number of users simultaneously.

◆ Data Integration

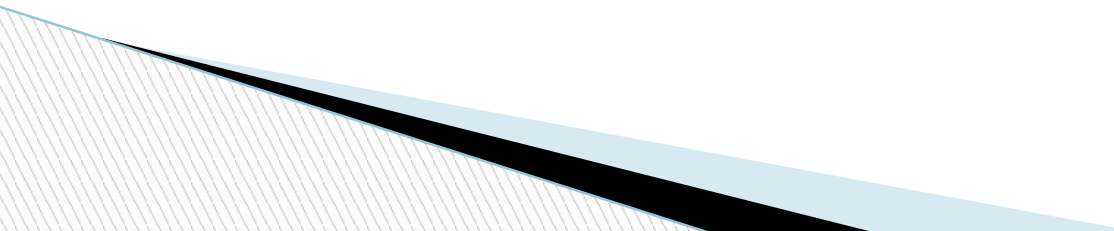
- In traditional file system, data is stored in separate files and it is difficult to extract the required information.
 - In database, data is stored in tables.
 - A single database may contain multiple tables.
 - The relationships can be created between tables .
 - This makes easier to retrieve and update data.
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Advantages of Database

◆ Data Integrity

- Data integrity refers to the correctness and consistency of data.
- It is expressed in terms of certain constraints.
- The consistency rules are applied on database to check that the correct data is entered in the database, before storing.

◆ Data Atomicity

- Atomicity means that either one transaction should take place as a whole or it should not take place at all.
 - If the any process is not completed successfully then the system fails and this is known as data atomicity problem.
 - In database partially completed tasks are rolled back.
 - Only consistent data exists within the database.
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Advantages of Database

◆ **Data Security**

- It is the protection of the database from unauthorized users.
- Only authorized persons are allowed to access the database.

◆ **Control Over Concurrency**

- In some situations two or more users may access the same file simultaneously.
- It is possible that they will interfere with each other.
- For example, two or more users are trying to update the same record, then one may overwrite the values recorded by other user. This may result in loss of information.
- Most databases control the concurrency so that transactions are recorded with accuracy.

Advantages of Database

9. **Better Backup and Recovery Facility:**

- Data is a very important resource and is very much valuable for any organization, loss of such a valuable resource can result in a huge strategic disasters.
- As Data is stored on today's' storage devices like hard disks etc., It is necessary to take periodic backups of data so that in case a storage device loses the data due to any damage we should be able to restore the data at nearest point.
- Database systems offer excellent facilities for taking backup of data and good mechanism of restoring those backups to get back the backed-up data.

Advantages of Database

◆ Data Independence

- The data stored in a database is independent of the application programs that accesses the data from database.
- The data structure of database and application program that uses data are separate from each other.
- The user can easily change the structure of database without modifying the application program.
- Similarly, user can modify application programs without changing structure of database.

Disadvantages of Database

1. High Cost:

Database includes the need for specialized **software** which is used to run database systems, Additional and specialized **hardware** and technically **qualified staff** are the requirements for adopting to the database system, all these requirements need an organization to invest handsome amount of money to have all the requirements of the database systems.

Disadvantages of Database

2. **Conversion Cost:** Once an organization has decided to adopt database system for its operations, it is not only the finance and technical man-power which is required for switching on to database system, it further has some conversion charges needed for adopting the database system, this is also a very important stage for making decision about the way the system will be converted to database system.

Disadvantages of Database

3. **Difficult Recovery Procedures:** Although the database systems and database management systems provide very efficient ways of data recovery in case of any disaster, still the process of recovering a crashed database is very much technical and needs good professional skills to perform a perfect recovery of the database.